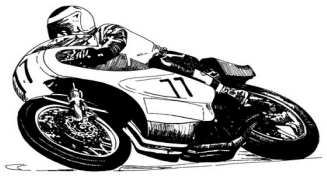
Year 10 - Forces and Motion Revision Notes

MOTION



Speed is how fast an object is moving

v_{av} is average speed of object in m/s (or units used for d & t) or $m.s^{-1}$

d is distance travelled in meters

t is time taken to travel distance in seconds

$v_{av} = d/t$

Example: Car travels at 800 $m.s^{-1}$ for 1 minute. How far did the car travel? *d= v x t*
d =800 x60 = 160m

Acceleration is the rate at which speed is changing

a is acceleration of object in $m.s^{-2}$

△ **v** is change in speed in m/s or $m.s^{-1}$

t = time taken to change speed in seconds

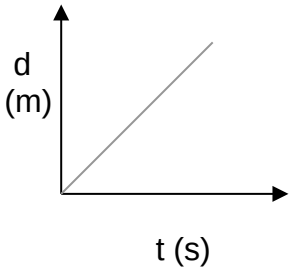
a = △v / t

Example: An object travelling at 25 $m.s^{-1}$ increases speed to 65 $m.s^{-1}$ in 5 seconds. What is the acceleration? *a = △ v / t* *a = 65 – 25 / 5 = 40 / 5 = 8 $m.s^{-2}$*

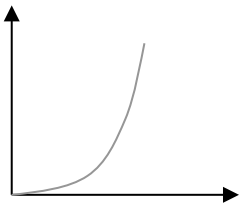
Distance – Time Graphs

Slope gives speed of object.

- Straight line = constant speed



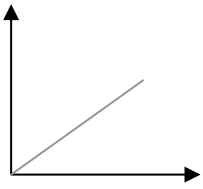
Curved line = increasing speed



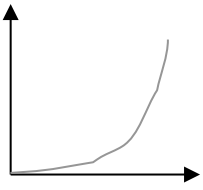
Speed – Time Graphs

Slope gives acceleration of object

- Straight line = constant acceleration



Curved line = increasing acceleration

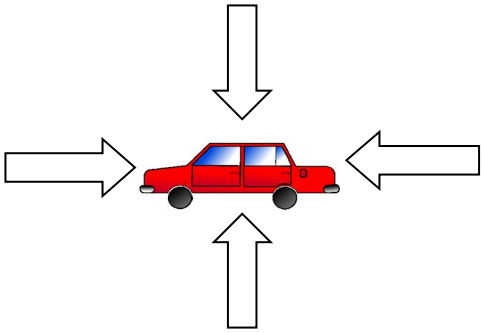


FORCES

- Forces are pushes or pulls.
- A force has a size and a direction, shown by an arrow.
- Symbol **F**
- Unit newton (N)
- **Net Force** is the combined effect of all forces on an object.
- Forces in the same direction add together.
- Forces in opposite directions subtract.
- **Weight force** is force due to gravity**.** **F = m x g** *g = 10 m.s^{-2}* *m = mass*
- **Friction force** opposes motion of object. Often causes heat.

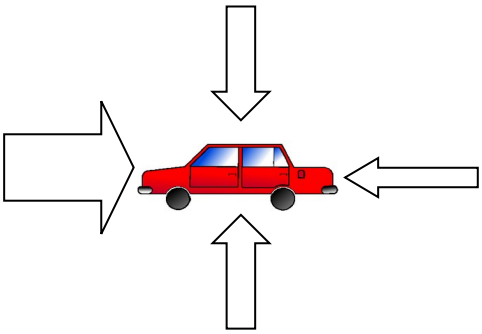
Balanced Forces

Combined effect (net force) is zero.
Object remains **stationary** or has **constant speed.**



Unbalanced Forces

Combined effect (net force) is not zero
Object **accelerates** in the direction of the net force



- **Calculate acceleration caused by unbalanced forces**

F = m x a

F is net force on object in newtons
m is mass of object in kilograms
a is acceleration on object in metres/sec/sec (m.s^{-2})

Calculate the acceleration of the above car if its mass is 1000 kg and net force acting is 250 N.
a = F/m *a = 250/1000* *a = 0.25 m.s^{-2}*

WORK AND POWER

Work

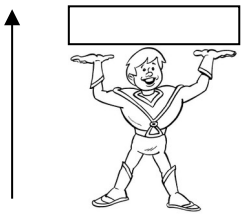
- Work is the amount of energy transferred when an object is moved over a distance.
- Work is only done when a force moves the object.

W **W = F x d** transferred to object, in joules

F is the force moving the object, in newtons
d is the distance object moves, in metres

Examples:

1. *A person pushes a toy car is pushed 2m across the floor with a force of 800 joules. Calculate the work done by the person.*
   W = F x d *W = 800 x 2 = 1600 J*
   The toy car has gained 1600 J of Kinetic Energy
2. *A weight lifter lifts a 50 kg box from ground to a height of 2m. What is the work done by the weightlifter?*
   The 50 kg box needs a force of 500N to lift it against gravity (F = m x g F = 50 x 10)
   W = F x d *W = 500 x 2 = 1000N*



The box has gained 1000N of Gravitational Potential Energy

Energy transfers above assume no energy lost to friction.

Power

- Power is the rate that work is done.
- Symbol **P**
- Unit watt (**W**)
- 1 watt means 1 joule of energy transferred (work done) each second.

P = W / t

P is power of machine in watts
W is work energy transferred to object, in joules
t is time taken to transfer energy, in seconds

Examples:

1. *Calculate the power output of the weightlifter above if box is lifted in 2 seconds.*
   P = W/t *P = 1000 / 2 = 500 W*
2. *Calculate power output of a crane that lifts 800kg of concrete a height of 10m in 20s.*
   F = m x g *F = 800 x 10 = 8000N*
   W = F x d *W = 8000N x 10 = 80000J*
   P = W/t *P = 80000 / 20 = 4000W*

Momentum

- Momentum is the tendency of objects to keep moving once they are in motion.
- Units $kg.m.s^{-1}$

momentum = m x v

m is mass, in kg **v** is objects speed, in m/s (ms^{-1})